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A data base for polymer chromatography: Dependence of interaction parameters on mobile phase composition

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ABSTRACT

In chromatography of polymers, retention is determined by the characteristic volumes of the column (pore volume and interstitial volume), the pore diameter, and the interaction parameter. While the influence of the pore diameter is predominant in size exclusion chromatography, the key parameters in liquid adsorption chromatography are the interaction parameter and the pore surface of the column. It is shown, that the retention behaviour of polymers in liquid adsorption chromatography (LAC) can be predicted very well using the accessible volume and pore surface of the column, which can be determined very easily, and the interaction parameters from a data base.

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1. Introduction

In chromatography of polymers, retention is determined by the characteristic volumes of the column (pore volume V_p and interstitial volume V_i), the pore diameter D , and the interaction parameter c . While the influence of the pore diameter is predominant in size exclusion chromatography (SEC), the key parameters describing retention in liquid adsorption chromatography (LAC) are the interaction parameter c and the pore surface S_p of the column, which is related to the pore volume V_p and pore diameter D .

In SEC, c is negative, in LAC it is positive. In **liquid chromatography under critical conditions (LCCC)**, $c=0$. If the parameters mentioned above are known with sufficient accuracy, one may predict the separation and even simulate chromatograms of very complex polymers [1–3]. The interaction parameter of a given repeat unit in LAC can be easily

determined by a procedure published in recent papers [4,5]. The interaction parameter is independent on column dimensions and pore diameter and can thus be used as a measure of the interaction of a given repeat unit with the surface of a stationary phase in a given mobile phase. In LAC, the dependence of the interaction parameter on mobile phase composition is typically assumed to be linear, but there are also other dependencies discussed [6]. In a plot of interaction parameters versus mobile phase composition ϕ , different columns with the same chemical nature should fall quite well on one straight line, the intercept of which represents the interaction parameter in the poor solvent (c_A), and the slope $dc/d\phi$ the strength of the good solvent. The intersection point of this line with the x-axis corresponds to critical conditions. If intercept and slope of these dependences (c_A and $dc/d\phi$) are known with sufficient accuracy, the interaction parameter of a given repeat unit in the LAC range

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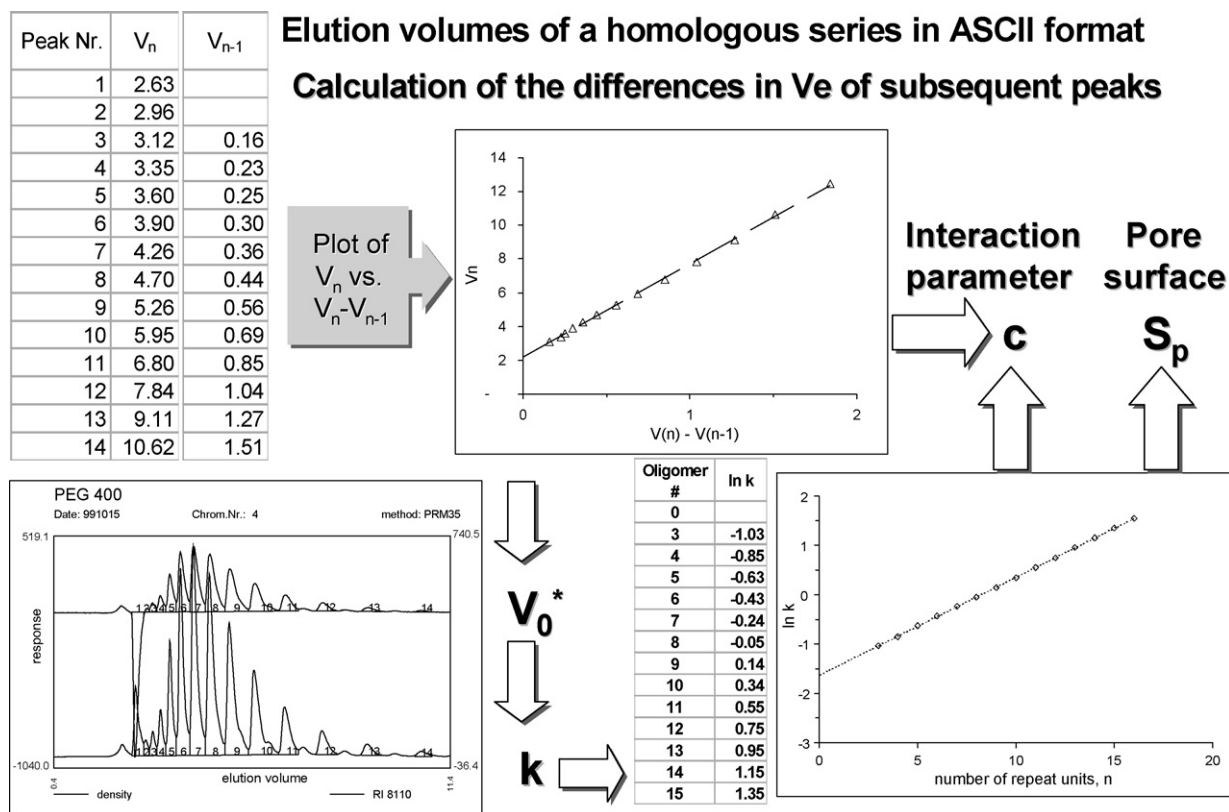


Fig. 1 – Determination of the interaction parameter and the pore surface (schematically).

can be calculated in any mobile phase composition for any column.

The retention of a given oligomer depends as well on the interaction parameter as on the pore dimensions (and thus the pore surface) of the stationary phase. As the interaction parameter of a given repeat unit in a given mobile phase is the same for stationary phases with the same chemical nature, retention can be adjusted by selection of the pore surface. This is especially important in the case of liquid chromatography under **critical conditions**, where retention of the invisible block cannot be adjusted by mobile phase composition or temperature [5,7,8].

The pore surface can be determined (in a similar way as the interaction parameter) from the elution volumes of non-functional oligomers [4,5,7]. The procedure is presented schematically in Fig. 1: if the elution volumes of a sufficient number of oligomers of the same (polymer) homologous series are known, they are plotted versus the difference $\Delta V_n = V_n - V_{n-1}$ in elution volumes of consecutive peaks.

$$V_n = V_0^* + \gamma \Delta V_n \quad (1)$$

The intercept in such a plot represents the accessible volume V_0^* . From the slope $\gamma = e^B / (e^B - 1)$ the interaction parameter c can be obtained: $B = (ca)^2 / 6$ is the slope in Martin's rule [9,10]

$$\ln k = A + Bn \quad (2)$$

and a is the length of the repeat unit.

It must be mentioned, that it is not necessary to know the number of repeat units for the individual peaks! Moreover, the interaction parameter for the repeat unit can be obtained regardless the end groups. Hence, one may use polymers with different functionality: as has already been shown in a previous paper [7], polyethylene glycols (PEG) and the corresponding mono- and dimethyl ethers fall on one straight line in such a plot. The only requirement is, that ΔV_n must be calculated for peaks with the same functionality (an example will be shown later on in Fig. 9).

If the peaks can be identified, the interaction parameter can be obtained from two other plots, as has been described previously [4,5,7]. One may plot the logarithm of the difference in elution volumes of consecutive peaks versus the number of repeat units n (Eq. (3)), or the logarithm of the retention factors (calculated with the accessible volume V_0^*) versus n (according to Martin's rule, Eq. (4)).

$$\ln \Delta V_n = \ln \frac{2S_p}{\gamma c} + Bn \quad (3)$$

$$\ln k^* = \ln \frac{2S_p}{cV_0^*} + n \frac{(ca)^2}{6} \quad (4)$$

In both cases, the slope yields the interaction parameter. For non-functional chains the pore surface S_p is obtained from the intercept of these plots.

Pore volume and thus pore surface may, however, depend on mobile phase composition: as we have shown recently [11], the pore volume of reversed phase columns in aque-

ous mobile phases can be considerably smaller than that in the pure organic solvent. Consequently, this approach is more relevant in LAC separations (which are run in mixed mobile phases) than the conventional method of determining the pore dimensions by inverse size exclusion chromatography (ISEC), which is typically performed in single mobile phase.

If the parameters mentioned above (c_A and $dc/d\phi$) for a given system of repeat unit, stationary phase and mobile phase are known from a data base, and individual columns can be characterized easily with respect to their accessible volume and pore surface, the retention of polymers can be predicted and even simulated for highly complex architecture by a special software [2,3,12], which even allows a simulation of two-dimensional chromatography. The same approach can also be applied to gradient chromatography, as will be shown in another paper.

In other words, such a data base would allow to select the starting conditions in method development much closer to the optimum, which can dramatically reduce time and solvent consumption required for the optimization of chromatographic conditions. In this paper, the requirements for such a data base should be evaluated by comparing the predicted results obtained using the parameters mentioned above with real chromatograms.

2. Experimental

These investigations were performed using the density detection system DDS70 (CHROMTECH, Graz, Austria). Data acquisition and processing were performed using the software package CHROMA, which has been developed for the DDS70.

In System A, the mobile phase was delivered by a JASCO 880 PU pump (from Japan Spectroscopic Company, Tokyo, Japan) at a flow rate of 0.5 mL min^{-1} .

Samples were injected manually using a Rheodyne 7125 injection valve (from Rheodyne, Cotati, CA, USA) equipped with an $50 \mu\text{L}$ loop. A Waters 2414 refractive index detector (Waters, Milford, MA, USA) was connected to the DDS70.

Columns were connected to two column selection valves (Rheodyne 7060, from Rheodyne, Cotati, CA, USA).

In system B, the mobile phase was delivered by an ISCO 2350 HPLC pump (ISCO, Lincoln, NE, USA). The flow rate was 0.5 mL min^{-1} in all measurements.

Samples were injected using a VICI injector equipped with an $50 \mu\text{L}$ sample loop. A SEDEX 45 ELSD (Sedere, France) was connected to the DDS70. Nitrogen was used as carrier gas, and the pressure at the nebulizer was set to 1.0 bar. The temperature of the evaporator was 30°C . The columns and density cells of both systems (A and B) were placed in a thermostated box, in which a temperature of 25.0°C was maintained for all measurements using a thermostat (Lauda RM6, Lauda-Königshofen, Germany).

The measurements in this study were performed on the following reversed phase columns (specifications given by the producer):

- Novapak C18 (Waters, Milford, MA, USA): silica-based octadecyl phase; $300 \text{ mm} \times 3.9 \text{ mm}$; particle diameter = $4 \mu\text{m}$; pore size = $60 \text{ \AA}$; surface area: $120 \text{ m}^2 \text{ g}^{-1}$; carbon load: 7%.

- Synergi $4 \mu\text{m}$ MAX-RP 80A (Phenomenex, Torrance, CA, USA): silica-based dodecyl phase; $250 \text{ mm} \times 4.6 \text{ mm}$; particle diameter = $4 \mu\text{m}$; pore size = $80 \text{ \AA}$; surface area: $475 \text{ m}^2 \text{ g}^{-1}$; carbon load: 17%.
- Prodigy ODS3 (Phenomenex, Torrance, CA, USA): silica-based octadecyl phase; $250 \text{ mm} \times 4.6 \text{ mm}$; particle diameter = $5 \mu\text{m}$; pore size = $100 \text{ \AA}$; surface area: $450 \text{ m}^2 \text{ g}^{-1}$; carbon load: 15.5%.
- Spherisorb S5X C18 (Phase Separations) silica-based octadecyl phase; $250 \text{ mm} \times 4.6 \text{ mm}$; particle diameter = $5 \mu\text{m}$; pore size = $300 \text{ \AA}$.
- Nucleosil 4000-7 C18 (Macherey-Nagel, Germany): silica-based octadecyl phase; $250 \text{ mm} \times 4 \text{ mm}$; particle diameter = $7 \mu\text{m}$; pore size = $4000 \text{ \AA}$.
- Chromolith Performance RP-18e 100-4.6 (Merck, Darmstadt, Germany): silica-based monolith with octadecyl groups; $100 \text{ mm} \times 4.6 \text{ mm}$; pore size: $130 \text{ \AA}$; surface area: $300 \text{ m}^2 \text{ g}^{-1}$; carbon load: 18%.
- Jordi Gel DVB 500 RP column (Jordi, Bellingham, MA, USA): 100% poly(divinyl benzene); $250 \text{ mm} \times 4.6 \text{ mm}$; particle diameter = $5 \mu\text{m}$; pore size = $500 \text{ \AA}$.

The solvents (methanol, acetone, THF and water, all HPLC grade) were purchased from Roth (Karlsruhe, Germany). Mobile phases were mixed by mass and vacuum degassed, their composition was controlled by density measurement using a DMA 60 density meter equipped with a measuring cell DMA 602M (A. Paar, Graz, Austria).

Fatty alcohols, polyethylene glycols (PEG) and polypropylene glycols (PPG) were purchased from Sigma-Aldrich.

3. Results and discussion

For prediction of chromatograms, the dependence of interaction parameters on mobile phase composition has to be known. Recently, Bashir and Radke [6] compared different models in LAC of PEG: the linear solvent strength model (LSSM), the quadratic solvent strength model (QSSM), both of which hold only in LAC, and the polymer model (PM), which can be applied in all modes of LC.

In previous papers, we have described the determination of the important parameters for PEG on various alkyl-bonded stationary phases in methanol–water and acetone–water mobile phases [4,5,7]. In this study, we have now applied the same approach to other repeat units and mobile phases. The interaction parameters were determined from the slope in a plot according to Eq. (1), as this does not require the unambiguous identification of the individual oligomers (with respect to n). In general, the agreement of the results obtained from Eqs. (1), (3) and (4) is very good.

As can be seen in Fig. 2, the interaction parameters of PPG on various alkyl-bonded stationary phases in acetone–water mobile phases show a linear dependence on mobile phase composition. The data obtained with different columns fall quite well on one straight line. For the quite old Spherisorb S5X C18 column we compared data obtained now and several years ago (1996–1999) and found, that there was no consid-

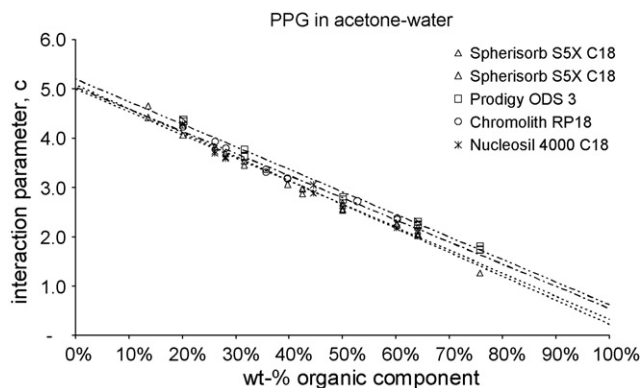


Fig. 2 – Interaction parameter of PPG on different reversed phase columns in acetone–water.

erable change in the interaction parameters over such a long period.

A similar behaviour was observed for fatty alcohols in methanol–water (Fig. 3): in this case, the repeat unit was the ethylene group.

A comparison of different samples (PEG, PPG and fatty alcohols) on the same column (Prodigy ODS3) in the same mobile phase (acetone–water) shows also reasonable straight lines with different slope and intercept (Fig. 4).

In these examples the dependence of interaction parameters on mobile phase composition appears to be linear, which means, that the linear solvent strength model describes retention quite well.

A linear dependence is also observed in a plot of the interaction parameters of PPG on Prodigy ODS3 in different mobile phases (Fig. 5): in the range covered by our measurements the data fall onto straight lines with a similar slope. On this column **critical conditions for PPG exist only in THF–water (at approximately 85 wt.% THF)**. The intersection point of the line for THF agrees well with the CAP, which means, that the dependence of interaction parameters on mobile phase composition is linear at THF concentrations higher than 30 wt.%.

One may expect, that all lines should have the same intercept (which represents the interaction parameter in pure water). In this plot, however, all lines have a different intercept. This suggests, that a considerable curvature must occur

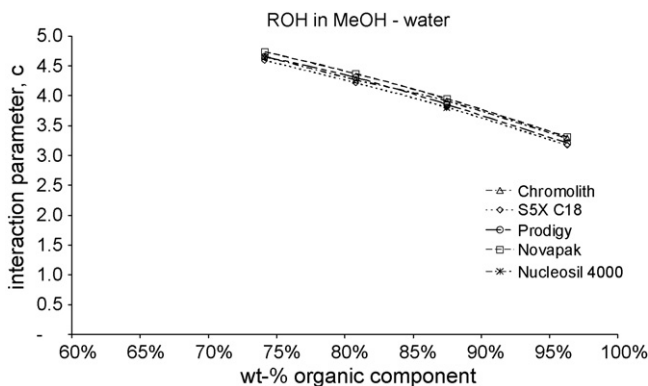


Fig. 3 – Interaction parameters of the ethylene unit on different reversed phase columns in methanol–water.

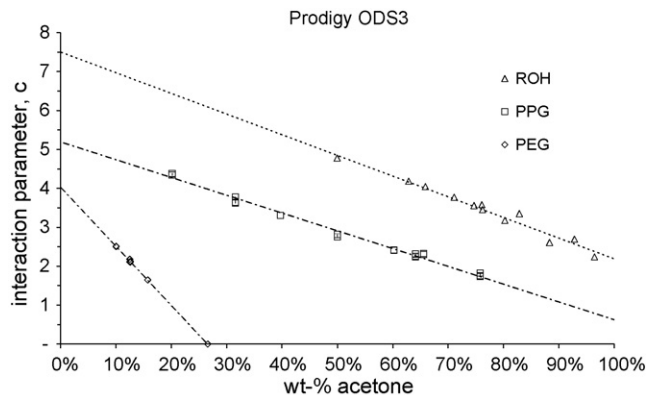


Fig. 4 – Interaction parameter of different repeat units on Prodigy ODS3 in acetone–water.

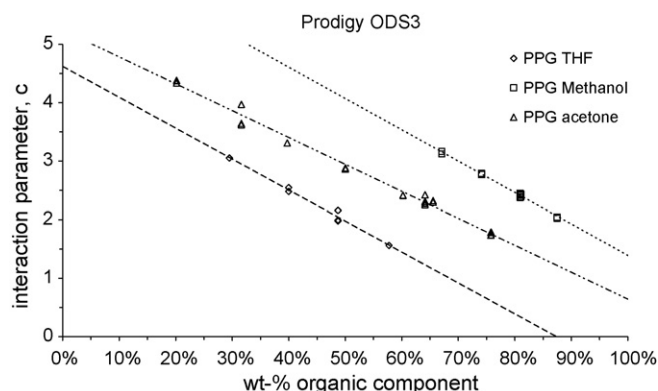


Fig. 5 – Interaction parameters of PPG on the Prodigy ODS3 column in different mobile phases.

at low concentrations of the organic solvent. The interaction parameter in mobile phases with a high content of water cannot be determined because of the poor solubility of PPG in such mobile phases. Consequently, these deviations are not relevant in practical application.

A different situation is observed for PEG (Fig. 6): the interaction parameters of PEG in acetone–water and methanol–water show a linear dependence for various columns in the range between c -values of 1.5 and 4 (which is actually the range, in which a reasonable separation by LAC is possible). Extrapolation to 0% organic component indicates the same interaction parameter in pure water, but near the critical point considerable deviations can be observed. The effect is even more pronounced on the Jordi column, which consists of 100% poly(divinyl benzene). While in acetone–water the results for PEG agree quite well with those obtained with alkyl-bonded stationary phases, there is no critical point at all in methanol–water mobile phases (Fig. 7). In this case, the best fit is obtained with the quadratic solvent strength model.

This means, that $dc/d\phi$ cannot be defined by the critical adsorption point (CAP) and a slope of the straight line near the CAP. Hence, it is highly desirable to cover the widest possible range of mobile phase composition. Obviously, the applicable range is rather limited on each column, retention depends, however, not only on the interaction parameter, but also on the pore surface.

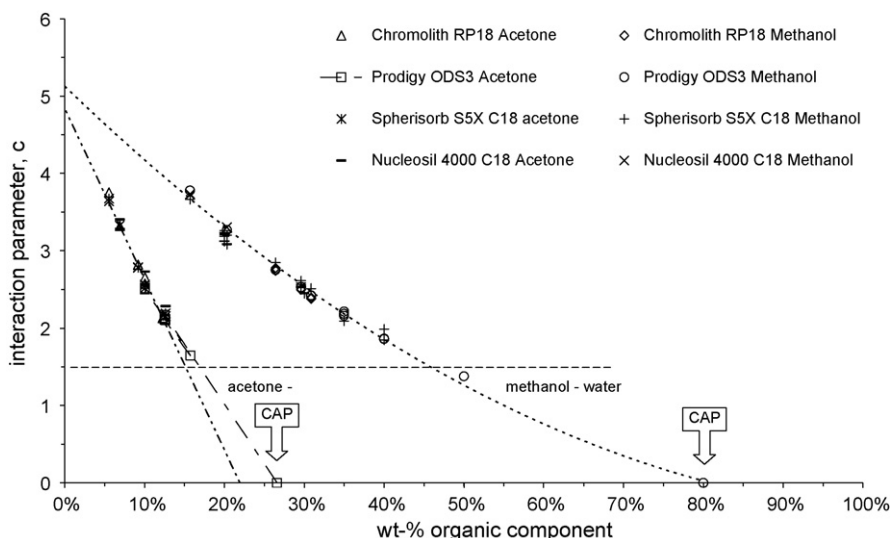


Fig. 6 – Interaction parameters of PEG on different columns in acetone–water and methanol–water mobile phases.

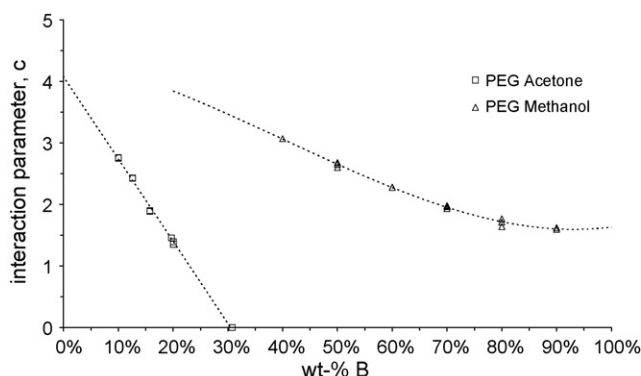


Fig. 7 – Interaction parameters of PEG on the Jordi column in acetone–water and methanol–water mobile phases.

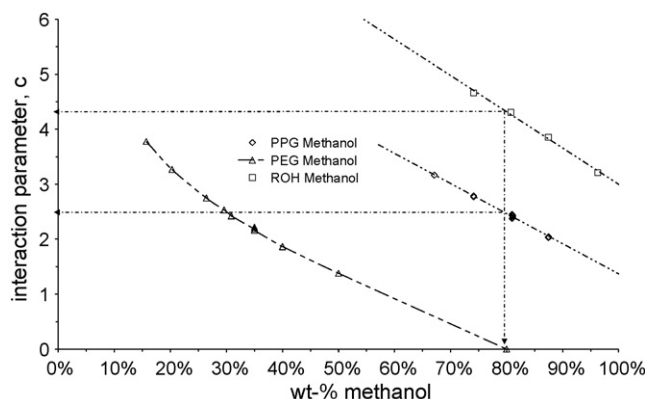


Fig. 8 – Interaction parameter of different repeat units on Prodigy ODS3 methanol–water.

Hence, the interaction parameter of a given repeat unit on stationary phases with the same chemical nature can be determined for a wide range of mobile phase composition by using columns with different pore surface.

Another important goal is the prediction of the retention of functional oligomers or block copolymers in LCCC: at the CAP for the repeat unit (the invisible block) the interaction parameter of the end group or the other block(s) determine the retention. As can be seen in Fig. 8, the interaction parameter of the ethylene unit at the CAP for the oxyethylene (EO) unit is much higher than that of the oxypropylene (PO) unit.

The final question concerns the predictive power of the new approach. As a test, we have compared real chromatograms with the elution volumes, which were calculated using the parameters c_A and $dc/d\phi$ obtained with other columns.

Fig. 9 shows two chromatograms of PEG 300 and PEG 400, which were obtained on the Synergi MAX RP column in 10.04 wt.% acetone. The PEG 400 shown here is an old sample, which obviously contains a second homologous series (maybe with oxidized end groups, peaks are indicated by arrows). These peaks must not be used together with the main series. If, however, only this series is used in the calculation, the inter-

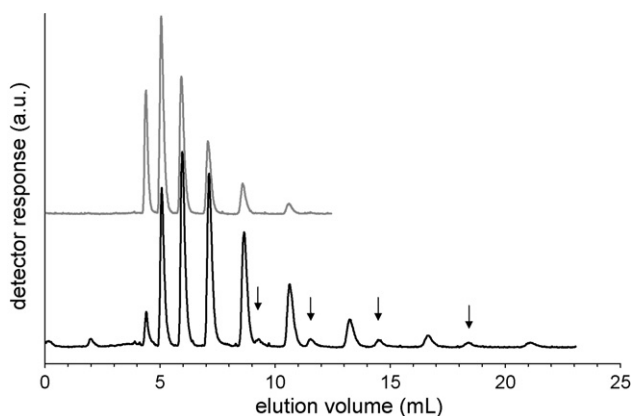


Fig. 9 – Chromatograms of PEG 300 and PEG 400 on the Synergi MAX RP column in 10.04 wt.% acetone. Detection ELSD.

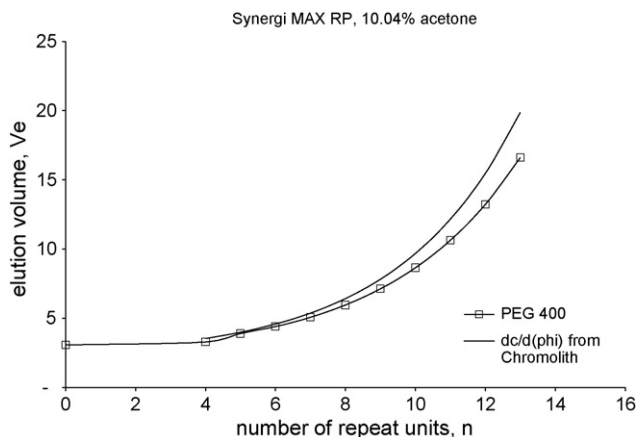


Fig. 10 – Elution volumes of PEG on the Synergi MAX RP column (Fig. 9) and calculated elution volumes using $d\phi/dt$ from the Chromolith column.

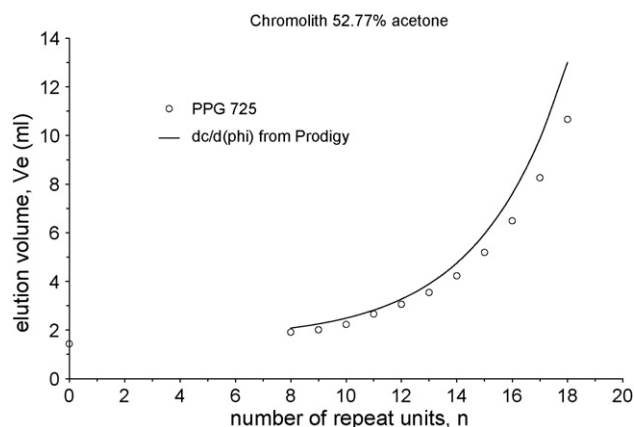


Fig. 11 – Elution volumes of PPG on the Chromolith RP 18 column and calculated elution volumes using $d\phi/dt$ from the Prodigy column.

action parameter for the EO unit agrees very well with that calculated from the main series.

The elution volumes of the individual oligomers were compared to those calculated with the parameters c_A and $dc/d\phi$ obtained with the Chromolith column. The accessible volume and the pore surface of the Synergi column were calculated from the chromatogram shown in Fig. 9. As can be seen in Fig. 10 the agreement of calculation and experimental data is quite good.

Another example is given in Fig. 11, which shows a comparison of elution volumes obtained for PPG 725 on the Chromolith column in 60.22 wt.% acetone with the calcu-

lated ones (using c_A and $dc/d\phi$ obtained with the Prodigy column).

4. Conclusions

The retention behaviour of polymers in LAC can be predicted very well using the accessible volume and pore surface of the column, which can be easily determined, and the interaction parameters from a data base. Such a data base can be established using data from isocratic LAC by a very simple procedure.

Existing data can be used: the only requirement is the knowledge of the elution volumes of a sufficient number of consecutive oligomers with the same functionality. Elution volumes of the individual oligomers in an isocratic LAC separation can be predicted with reasonable accuracy using interaction parameters from a data base (which have been determined using other columns with the same chemical nature) and the characteristic parameters for the column used (accessible volume and pore surface), which could be determined in principle by one single measurement. In practice, three to five chromatograms will be satisfactory to obtain these parameters with good accuracy. In the range of interaction parameters typically used in LAC (c between 2.0 and 5.0) simple equations can be used to describe retention of homopolymers. The same approach can also be applied in gradient separations, as will be shown in another communication.

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